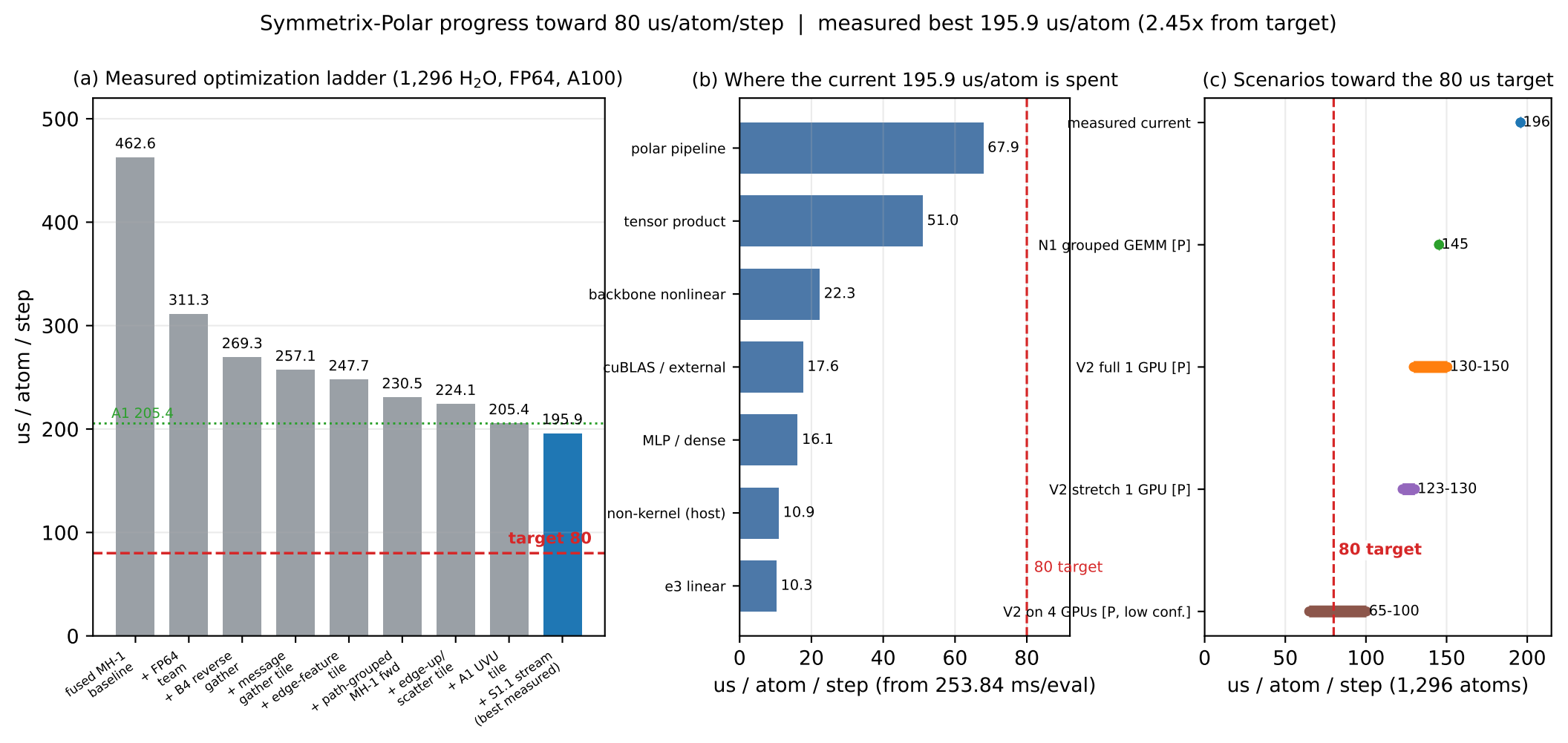
Symmetrix-Polar progress toward 80 us/atom/step | measured best 195.9 us/atom (2.45x from target)
(a) Measured optimization ladder (1,296 H2O, FP64, A100)
us / atom / step
462.6
311.3
269.3
257.1
247.7
230.5
224.1
205.4
195.9
A1 205.4
target 80
fused MH-1 baseline
+ FP64 team
+ B4 reverse gather
+ message gather tile
+ edge-feature tile
+ path-grouped MH-1 fwd
+ edge-up/ scatter tile
+ A1 UVU tile
+ S1.1 stream (best measured)
(b) Where the current 195.9 us/atom is spent
polar pipeline
67.9
tensor product
51.0
backbone nonlinear
22.3
cuBLAS / external
17.6
MLP / dense
16.1
non-kernel (host)
10.9
e3 linear
10.3
80 target
us / atom / step (from 253.84 ms/eval)
(c) Scenarios toward the 80 us target
measured current
196
N1 grouped GEMM [P]
145
V2 full 1 GPU [P]
130-150
V2 stretch 1 GPU [P]
123-130
V2 on 4 GPUs [P, low conf.]
65-100
80 target
us / atom / step (1,296 atoms)